SOFTWARE ENGINEERING LAB – 2

Agile Backlog Creation & Sprint Simulation in Jira

Pharmacy Expiry Reorder System

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Problem Statement	#15 – Pharmacy Expiry & Re-order Dispatch Engine
Domain	Healthcare & Telemedicine
Jira Project	Pharmacy Expiry Reorder System (PERS)

This report documents the Jira backlog, Agile user stories, story-point estimates, sprint simulation, burndown evidence, and reflection for the assigned problem statement.

1. EPICS

The lab instructs students to break Lab 1 requirements into major functional themes (Epics) before creating User Stories.

Epic	Description
Medicine Batch & Expiry Management	Manage medicine batches, batch numbers, and expiry dates to ensure accurate tracking a
Inventory & Stock Monitoring	Maintain accurate medicine stock quantities and monitor inventory levels against configure
Automated Re-order Management	Automatically generate and dispatch purchase orders to inventory suppliers when medicin

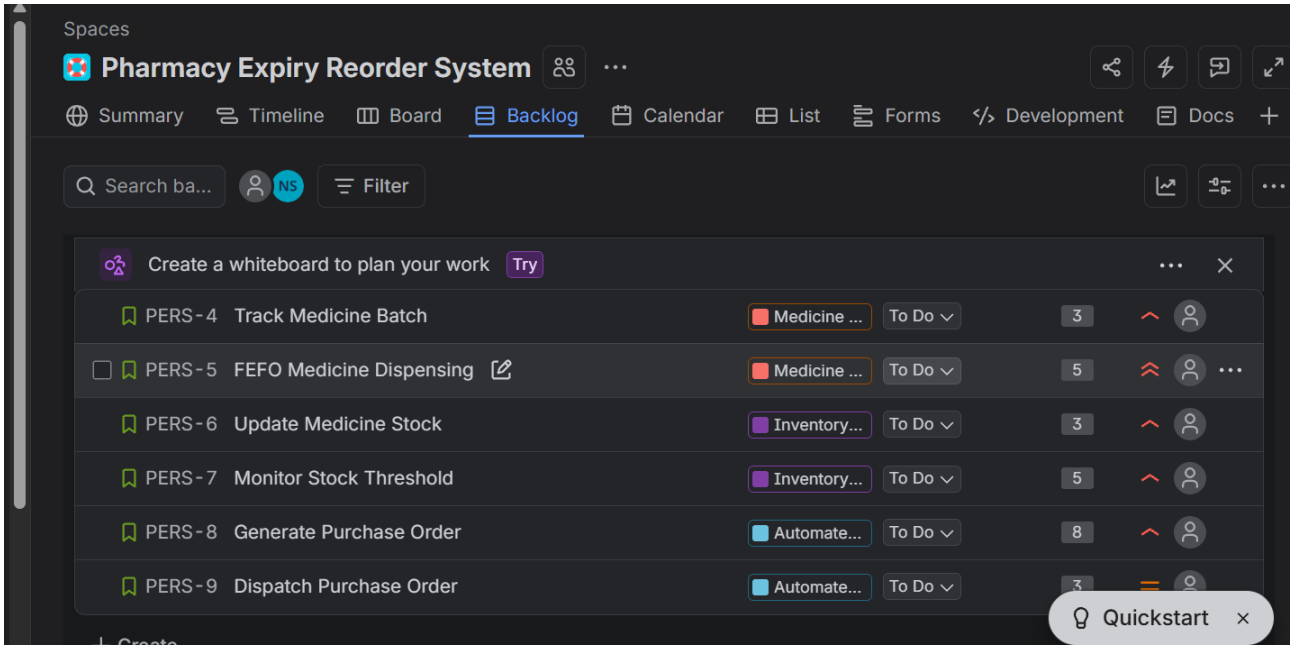


Figure 1. Jira backlog showing the six User Stories, their Epic associations, and story points.

2. USER STORIES, PRIORITY & STORY POINTS

The lab uses the format As a [user role], I want [goal], so that [benefit]. Story points are estimated using the Fibonacci-style sequence.

#	User Story	Priority	Points	Description
1	Track Medicine Batch	High	3	As a Pharmacy Clerk, I want to record medicine batch numbers and expiry dates, so that I can ac
2	FEFO Medicine Dispensing	Highest	5	As a Pharmacy Clerk, I want the system to select the medicine batch with the nearest valid expir
3	Update Medicine Stock	High	3	As a Pharmacy Clerk, I want the system to automatically update medicine quantities after dispen
4	Monitor Stock Threshold	High	5	As a Pharmacy Clerk, I want the system to monitor medicine stock levels against safety threshol
5	Generate Purchase Order	High	8	As a Pharmacy Clerk, I want the system to automatically generate a purchase order when medic
6	Dispatch Purchase Order	Medium	3	As an Inventory Supplier, I want to receive purchase orders from the pharmacy system, so that I

Total backlog estimate: 27 story points.

3. SPRINT 1 – ACTIVE SPRINT EVIDENCE

Sprint: PERS Sprint 1 **Duration shown by Jira:** 5 Oct 2026 – 12 Oct 2026

Workflow: To Do → In Progress → Done

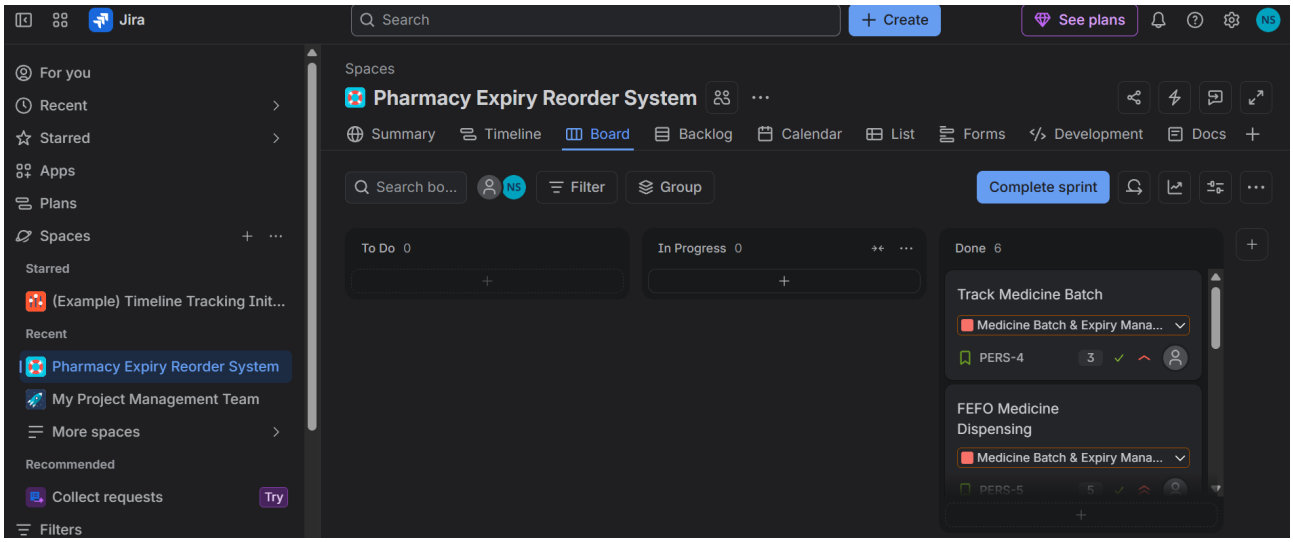


Figure 2. Jira Active Sprint board showing completed work items.

4. SPRINT 1 BURNDOWN CHART

Jira's Sprint 1 report uses Story points as the estimation field. The chart shows remaining work and the ideal guideline.



Figure 3. PERS Sprint 1 Burndown Chart.

5. SPRINT 2 – ACTIVE SPRINT EVIDENCE

Sprint: PERS Sprint 2 **Work item:** Dispatch Purchase Order **Points:** 3

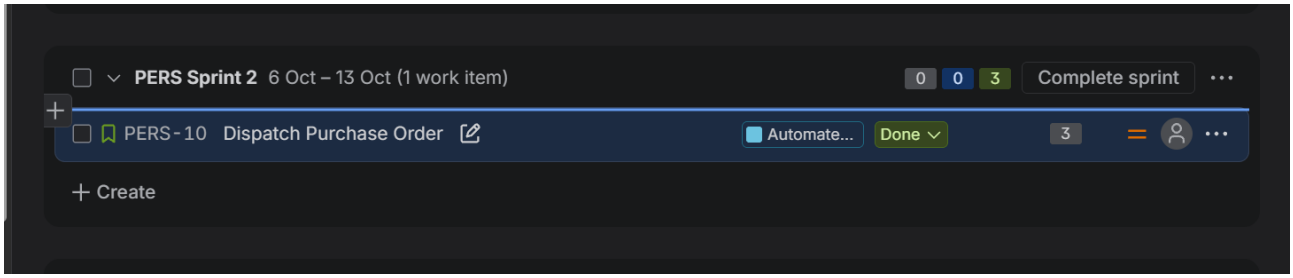


Figure 4. PERS Sprint 2 showing Dispatch Purchase Order completed.

The Lab 2 handout instructs students to finish two sprints; Sprint 2 contains the remaining 3-point work item.

6. SPRINT 2 BURNDOWN CHART

Jira's Sprint 2 report shows Story points as the estimation field and the 3-point work item reaching zero remaining work.



Figure 5. PERS Sprint 2 Burndown Chart.

7. REFLECTION

1. Did your estimations reflect the actual effort?

Yes. The story points reasonably reflected the relative effort and complexity of the user stories. Simpler activities such as updating medicine stock received fewer points, while generating a purchase order received more points because it involves additional processing and supplier-related functionality.

2. Was your backlog well-prioritized?

Yes. Important pharmacy operations such as FEFO dispensing, stock updating, stock monitoring, and purchase-order generation were given high priority because they directly affect medicine availability and inventory management.

3. How did your simulated sprint align with your plan?

The simulated sprint followed the planned Agile workflow. Work items were moved through the Jira status flow and the remaining work item was completed in Sprint 2, demonstrating the sprint-based approach.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts showed remaining story points during each sprint. The work reached zero after the planned items were completed, indicating that the simulated sprint work was completed successfully.

8. FINAL CHECKLIST

Deliverable	Status
Epics	Completed
User Stories	6 created
Priority assignments	Completed
Story points	Completed
Sprint 1	Completed
Sprint 2	Completed
Active sprint evidence	Included
Sprint 1 Burndown Chart	Included
Sprint 2 Burndown Chart	Included
Reflection questions	Answered

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